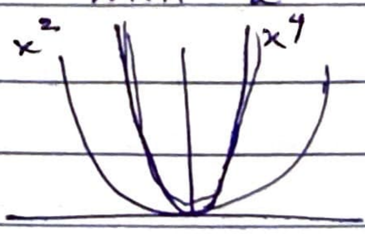
23rd Feb 2026 | DA6360

⇒ Newton vs GD (contd.)

- $\min x^4$



$$f'(x) = 4x^3$$
$$f''(x) = 12x^2$$

Newton

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$
$$= x_k - \frac{4x_k^3}{12x_k^2}$$
$$= x_k - \frac{1}{3}x_k$$
$$x_{k+1} = \underbrace{\frac{2}{3}}_{\text{shrinkage factor}} \cdot x_k$$
$$x_k \to x_* = 0$$

GD

$$x_{k+1} = x_k - \alpha 4x_k^3$$
$$= \underbrace{(1 - \alpha 4x_k^2)}_{\text{shrinkage factor}} x_k$$

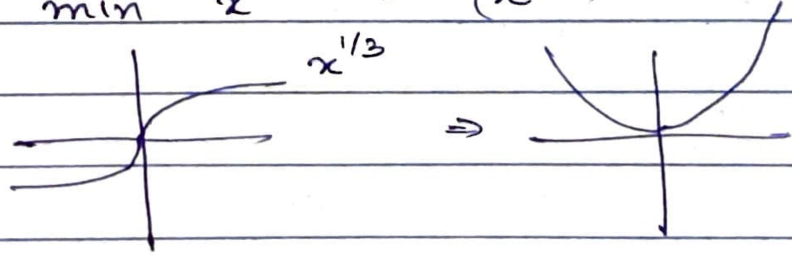
Suppose, $x_k \to 0$ (getting close to)

$\therefore, x_k^2 \to 0$

$\therefore, (1 - \alpha 4x_k^2) \to 1$

$$x_{k+1} \simeq 1 \cdot x_k$$

- $\min x^{4/3} = (x^{1/3})^4$



$$f'(x) = \frac{4}{3}x^{1/3}$$
$$f''(x) = \frac{4}{9}x^{-2/3}$$

Newton

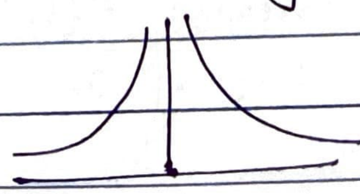
$$x_{k+1} = x_k - \frac{4/3\, x^{1/3}}{4/9\, x^{-2/3}} = x_k - 3x_k = -2x_k$$

$$x_{k+1} = \underbrace{-2}_{\text{does not shrink}} x_k$$

$$x_{k+1} = x_k - \alpha \frac{4}{3} x_k^{1/3}$$

This means
the given func^n can't converge.

$$f''(x) = \frac{4}{9} \frac{1}{x^{2/3}}$$



at any given time, we feel that slope is fine, but $f''(x)$ shows that the slope is much steeper. The estimate in $f''(x)$ is not that good! In the earlier eg., the slope was much slower.

⇒ **Rate & Order of Convergence**

- Order of convergence & rate of convergence

if $\{x_k\} \to x_*$, $e_k = x_k - x_*$

$$\boxed{\lim_{k \to \infty} \frac{|e_{k+1}|}{\underbrace{|e_k|^q}_{\text{order}}} = \mu} \to \text{rate}$$

how many decimal places we progress

$q = 1$, $\mu = 0.1$, $e_0 = 1$

$|e_1| = 0.1$

$|e_2| = 0.02$

$q = 2$, $\mu = 9$, $|e_0| = 0.1$

$|e_1| = 0.09$, $|e_2| = 0.0729$ ...

that means the newton method has an order of 2. $(q = 2)$

- 3rd order approximaⁿ (justificaⁿ)

$$f(x) \simeq f(x_k) + f'(x_k)(x - x_k) + f''(x_k)\frac{(x-x_k)^2}{2!} + f'''(x_k)\frac{(x-x_k)^3}{3!}$$

$$f(x) = \; - \; + \; \_\_\_ \; + f'''(z)\frac{(x-x_k)^3}{3!}$$

$$f'(x) = 0 + f'(x_k) + f''(x_k)(x - x_k) + f'''(z)\frac{(x-x_k)^2}{2!}$$

we have to use the fact that x_*, $f'(x_*) = 0$

$$0 = 0 + f'(x_k) + f''(x_k)(x_* - x_k) + f'''(z)\frac{(x_* - x_k)^2}{2!}$$

$$-\frac{f'(x_k)}{f''(x_k)} = x_* - x_k + \frac{f'''(z)}{f''(x_k)}(x_* - x_k)^2$$

$$x_k - \frac{f'(x_k)}{f''(x_k)} - x_* = \frac{1}{2}\frac{f'''(z)}{f''(x_k)}(x_* - x_k)^2$$

$$x_{k+1} - x_* = \frac{1}{2}\frac{f'''(z)}{f''(x_k)}(x_k - x_*)^2$$

$$|e_{k+1}| = \frac{1}{2}\left|\frac{f'''(z)}{f''(x_k)}\right| |e_k|^2$$

Case I

$$\boxed{\frac{|e_{k+1}|}{|e_k|^2} = \frac{1}{2}\left|\frac{f'''(z)}{f''(x_k)}\right|}$$

if $\frac{f'''(z)}{f''(x_k)} < 1$ this shrinks, then good

Case II

$$\frac{|e_{k+1}|}{|e_k|} = \underbrace{\frac{1}{2}\left|\frac{f'''(z)}{f''(x_k)}\right| |e_k|}$$

if at least this < 1, then also fine.

if not case I & II, then no convergence is possible.

eq. $f(x) = ax^2 + bx^4$

$$\frac{1}{2}\left(\frac{f'''(z)}{f''(x_k)}\right) = \cancel{\frac{1}{2}\left(\frac{2az + 4bz^3}{2a + 12bx_k^2}\right)}$$

$$\cancel{= \frac{1}{2}\left(\frac{a + 2bz^2}{2a + 12bx_k^2}\right) z}$$

$$= \frac{1}{2}\left(\frac{24bz}{2a + 12bx_k^2}\right) < \left|\frac{12bz}{12bx_k^2}\right|$$

$$= \frac{1}{2}\left|\frac{12bz}{2a + 12bx_k^2}\right| < \left|\frac{z}{x_k^2}\right|$$

if this goes to 0.
$2a$ is constant
& provides a tighter bound.

eq. $f(x) = x^4$

$$\frac{1}{2}\left(\frac{f'''(z)}{f''(x_k)}\right) = \frac{1}{2}\left|\frac{24z}{12x_k^2}\right| = \frac{z}{|x_k|^2} \to \infty$$

(failed)

eq. $f(x) = x^{4/3}$ $\Rightarrow$ does not even converge

$$f'''(x) = \frac{-8}{27} x^{-5/3}$$

$$\frac{1}{2}\left|\frac{8/27 \, z^{-5/3}}{\frac{4}{9} x_k^{-2/3}}\right| = \frac{1}{2}\left|\frac{8/27 \, x_k^{+2/3}}{4/9 \, z^{+5/3}}\right|$$

$$= \frac{1}{3}\frac{x_k^{2/3}}{z^{5/3}} \longrightarrow \infty$$

$z^{5/3} \Rightarrow$ much smaller

linear rate also fails.